

PO Co-Pilot Writeup

Problem

Manual data entry is still heavily relied on in the New Zealand IT distribution/reseller industry. Placing an order as a reseller usually involves sending a PO PDF email that includes all the standard information along with special pricing information and delivery instructions. The distributor must then enter this data into their ERP system to process the order. Doing this process manually requires a full team. It's time-consuming and very error-prone. These inevitable errors cause many negative flow-on effects that cost time and money to fix.

User

Order Management teams are responsible for entering the relevant data into the ERP system so the orders can be fulfilled. Depending on the size of a distributor, the team could be responsible for a high volume of orders. Members of the team will have other tasks, such as follow-up communication with customers and fixing any errors and issues that come up.

Current Experience

Currently, members of the team receive POs via email and must enter the relevant information into the ERP system. The essential information would be:

- PO Reference
- Delivery address and Contact Information
- SKUs
- Quantities and Unit Prices

There may be additional information such as bulk rate charges, special pricing, and delivery instructions also. This information must be entered correctly for the order to be completed successfully. The main method of doing that is copying and pasting from the PDF/email to the ERP system. Automating the data entry of POs will free up time and focus on quality assurance in dealing with the issues and customer requests.

Solution

A command line tool that reads a PO PDF or email and extracts the essential information needed and creates a CSV file as the output. The information can be manually entered into the ERP system in the order that it is required. The CSV file outputs the data in the correct order. The PDFs and emails are placed in separate folders. The tool reads each one and creates a separate CSV file for each order.

The second tool is an Email Triage Agent that can read a folder of emails and identify the ones that contain POs that need to be processed. It uses an Anthropic API key to classify each email, pushing PO/PDFs to the command line tool.

Demo

Example input:

Harborline Engineering Ltd 12 Apirana Avenue Glen Innes, Auckland 1072 Phone: (09) 555 4821 Fax: (09) 555 4820 Website: www.harborline-eng.co.nz		PURCHASE ORDER												
		DATE 02/05/2026 PO # PO-2026-04812												
VENDOR Pacific Fastener Supplies Ltd Trade Counter / Accounts 47 Patiki Road Avondale, Auckland 1026 Phone: (09) 828 7733 Fax: (09) 828 7734		SHIP TO Toby Fitzgerald Harborline Engineering Ltd 12 Apirana Avenue Glen Innes, Auckland 1072 (09) 555 4821												
REQUISITIONER	SHIP VIA	F.O.B.	SHIPPING TERMS											
M. Tanaka	Mainfreight Road	Destination	Net 30 days											
ITEM #	DESCRIPTION	QTY	UNIT PRICE	TOTAL										
FX-M08-50-SS	M8 x 50mm hex bolt, 316 stainless steel	250	0.84	210.00										
FX-NUT-M08-ZN	M8 hex nut, zinc plated, DIN 934	500	0.18	90.00										
FX-WSH-M08-FL	M8 flat washer, zinc, 25mm OD	500	0.09	45.00										
FX-RVT-48-AL	Pop rivet 4.8 x 12mm aluminium (box of 500)	4	38.50	154.00										
AB-LOC-243-50	Threadlock medium-strength, 50ml bottle	6	24.75	148.50										
LU-WD40-400	WD-40 multi-use 400g aerosol	12	12.40	148.80										
Comments or Special Instructions Please deliver between 8:00 and 16:00 weekdays. All goods to be tagged with PO number on packing slip. Stainless items must include mill cert. If you have any questions about this purchase order, please contact Toby Fitzgerald, (09) 555 4821, toby.fitzgerald@harborline-eng.co.nz		<table border="1"> <tr> <td>SUBTOTAL</td> <td>796.30</td> </tr> <tr> <td>GST 15%</td> <td>119.44</td> </tr> <tr> <td>SHIPPING</td> <td>28.00</td> </tr> <tr> <td>OTHER</td> <td>0.00</td> </tr> <tr> <td>TOTAL</td> <td>NZ\$ 943.74</td> </tr> </table>			SUBTOTAL	796.30	GST 15%	119.44	SHIPPING	28.00	OTHER	0.00	TOTAL	NZ\$ 943.74
SUBTOTAL	796.30													
GST 15%	119.44													
SHIPPING	28.00													
OTHER	0.00													
TOTAL	NZ\$ 943.74													

Example output:

PO-2026-04812

PO Number					
PO-2026-04812					
Company Name	Street Address	Suburb	Postcode	Contact Name	Phone
Harborline Engineering Ltd	12 Apirana Avenue	Glen Innes	1072	Toby Fitzgerald	(09) 555 4821
SKU	SKU Qty	Unit Price			
FX-M08-50-SS	250	0.84			
FX-NUT-M08-ZN	500	0.18			
FX-WSH-M08-FL	500	0.09			
FX-RVT-48-AL	4	38.50			
AB-LOC-243-50	6	24.75			
LU-WD40-400	12	12.40			

Tech

The tech used is a PDF Plumber. Python 3.9 and Anthropic API key were chosen because they were easy to set up and use on Mac OS. PDF Plumber has most of the required functionality built in already. Python is installed on Mac OS. The code was written using Claude Code and a Claude chat.

What's Next

The next step would be an API integration with the ERP system to completely automate the data entry process. Combined with the email triage agent, a PO could be sent to a mailbox and entered into the ERP system without any human intervention.